

BCBS 239 Principles & Data Governance Implementation

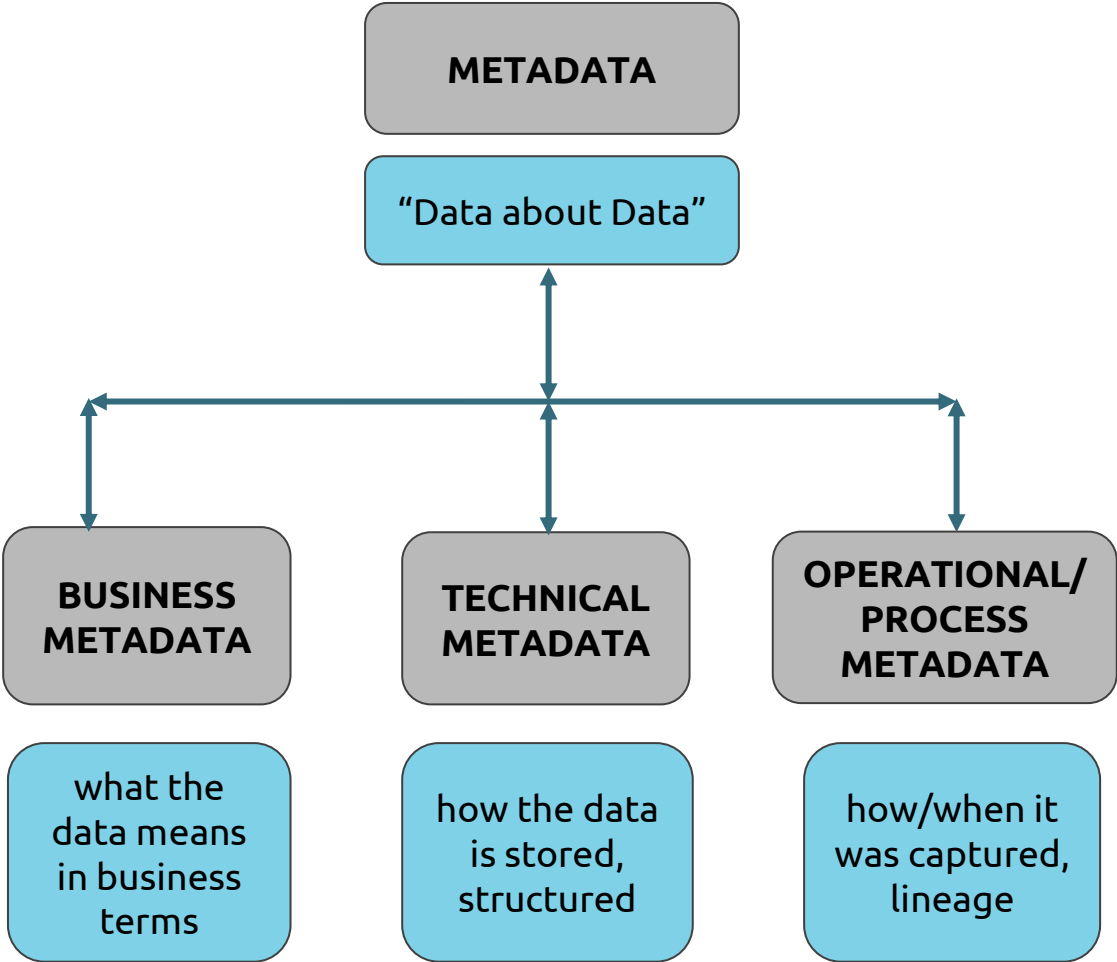
Knowledge Sharing Session



BCBS 239

- 1 Quick summary of What we already know
- 2 GSIB, DSIB and Compliance Assessment
- 3 Implementation of capabilities**
- 4 Real world examples
- 5 Open Discussion

Metadata Management



Key Components	
Metadata Management	- Governance process for capturing & maintaining metadata - Enables lineage tracking - Integrates business, technical, operational metadata
Core Tools	Data Dictionary: Defines names, types, valid values Business Glossary: Common business terms, owners, synonyms
Why It Matters	- Discover data: organizes data into searchable, well-organized, and accessible information assets. - Understand Data relationships : enables users to visualize lineage, dependencies, and data transformations end to end. - Track usage : captures who uses the data, when, and for what purpose. - Assess value & risk: allows organizations to evaluate which data is critical, sensitive, or high-value.

Types of Metadata



Descriptive metadata

Descriptive metadata includes basic information, **such as title, author, keywords and summaries**. This type of metadata helps organizations improve the searchability and discoverability of their data in [catalogs](#), social media platforms and search engines.

Structural metadata

Structural metadata describes and defines **how data elements are organized and interrelated**. For example, how a homepage links to subpages. This type of metadata helps organizations maintain clear relationships and categorizations within complex datasets.

Administrative metadata

Administrative metadata encompasses **ownership, permissions and retention policies**. This type of metadata helps organizations comply with legal, regulatory and internal policies. It defines data usage policies such as who can access data and how long it should be retained.

Technical metadata

Technical metadata comprises a data file's technical properties, like **format, encoding and storage location** (such as [data warehouses](#) or [data lakes](#)). This type of metadata helps organizations handle and display data correctly across different platforms and systems.

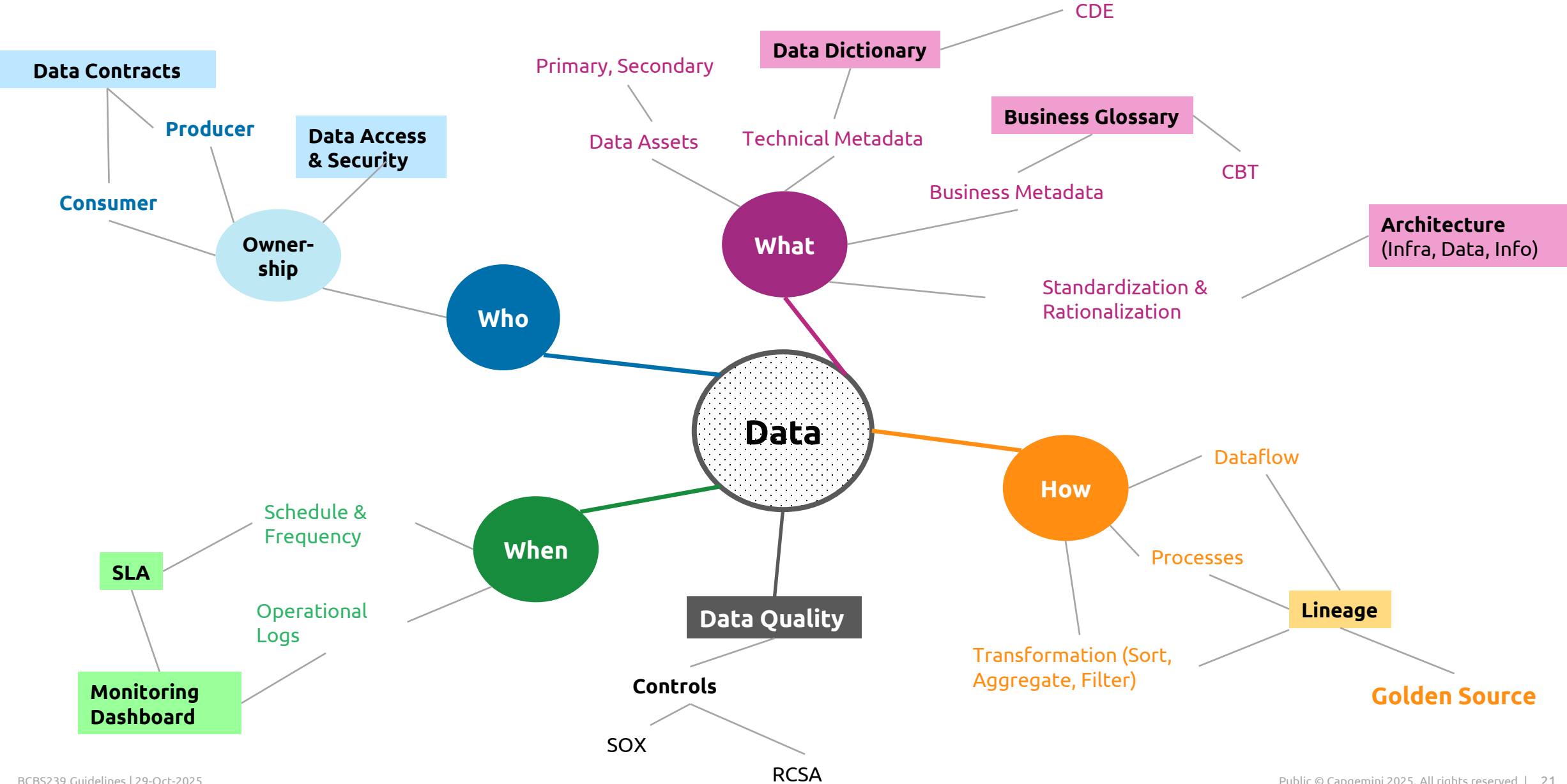
Preservation metadata

Preservation metadata ensures the long-term usability and accessibility of data, including **strategies for data backups and migration to newer formats**. This type of metadata helps organizations meet extended **data-retention requirements**, especially in industries like healthcare and legal services where records must remain accessible for compliance.



Capability	Collibra	Atlan	Informatica	Alation	OEMM	Atlas	DataHub
Business-term catalogue	✓	✓	✓	✓	⦿	✓	⦿
Central data dictionary	✓	✓	✓	⦿	✓	✓	✓
End-to-end lineage view	✓	✓	✓	✓	✓	✓	✓
Data-quality linkage	✓	⦿	✓	⦿	⦿	⦿	⦿
Governance workflow	✓	✓	✓	✓	⦿	⦿	⦿
Automated metadata capture	✓	✓	⦿	⦿	✓	✓	✓
User collaboration layer	✓	✓	⦿	✓	⦿	⦿	⦿
Open-API / extensibility	⦿	✓	⦿	⦿	✓	✓	✓
Cloud or hybrid deployment	✓	✓	✓	✓	⦿	✓	✓

Metadata Management Framework - Mind Map





- Data dictionary tools help organize and document data by providing a central repository for metadata, definitions, and relationships.

Database	Schema	Table Name / File Name	Attribute Name	Attribute Display Name	Description	Data Type	PII Flag	Sample Records	Comments & Additional Information
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Tools

- Popular options include commercial products like **Alation, Collibra, and Secoda, Dataedo** as well as open-source tools such as **DataHub, and Apache Atlas, Amundsen etc.**
- Many database, ETL, Data modelling systems have built-in or integrated tools – **IBM DB2, Oracle DB, SQL Server Management Studio, Informatica, Erwin Modeler etc.**

Most of the tools support **active dictionary** feature, that eliminates manual efforts to create and maintain data dictionary

Business Glossary



A business glossary is a **centralized collection of standardized terms and definitions** used within an organization to ensure everyone uses the same language and understands key concepts consistently

(CBT) Contextual Business Term	(BT) Bank Specific Business Term	Sub Domain	Entity / Geography	Is Critical?	Information Classification	PII Flag	Created by	Approver	Mapped CDE
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Following are few of the well-known tools in the market for Business Glossary

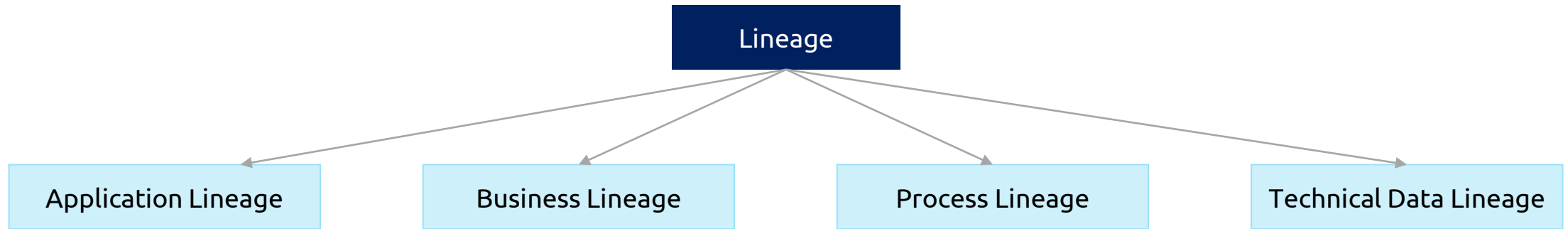
- Collibra
- Informatica Enterprise Data Catalog
- Talend Data Catalog
- Alteryx Connect
- IBM InfoSphere Information Governance Catalog
- Erwin
- Apache Atlas
- Alation Data Catalog
- Oracle Enterprise Metadata Management

There is a high potential for **Gen AI** in **“Business glossary rationalization”** exercise in large Banks having complex products and org structure

Lineage & Golden Source



Different types are lineage are documented to establish traceability of data transformations, business terms and golden source

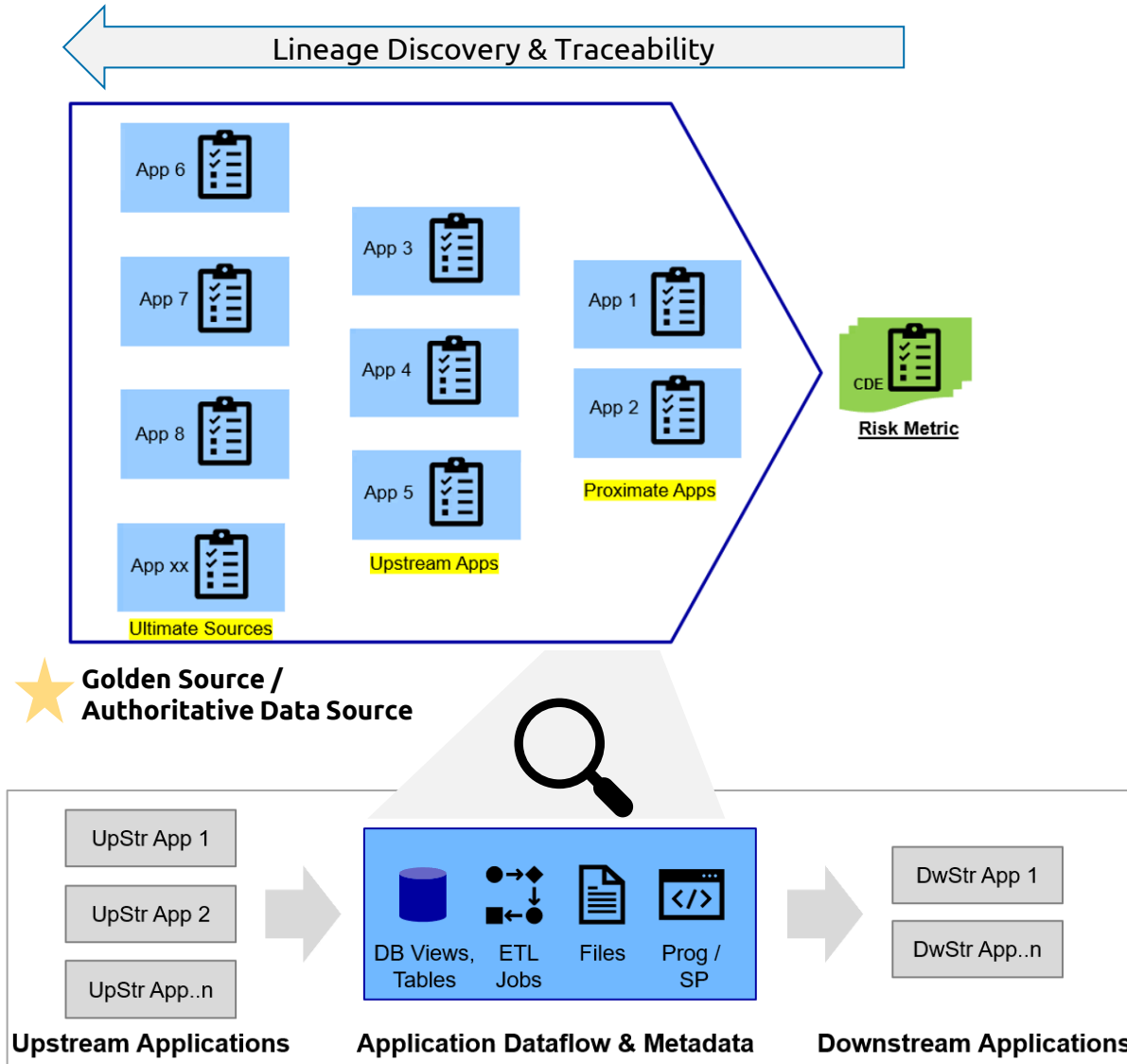


Data Lineage & Golden Source



	APPLICATION LINEAGE	BUSINESS LINEAGE	PROCESS LINEAGE	TECHNICAL LINEAGE
Importance	Which systems are involved	Why the data matters	Where the data is used in workflows	How the data moves
End Users	Application owners, IT governance, enterprise architects.	Business Analysts, Business Owners, Data Stewards	Process owners, auditors	Data Engineers, Data Architects, IT teams
Details	Shows which applications produce, consume, or modify data. Helps understand dependencies between systems	Provides a summary view of how data flows from its source to where it is consumed	Maps data to business workflows (e.g., KYC onboarding, credit approval, suspicious activity monitoring)	More granular and gives data engineers and other technical users a zoomed-in view of technical dataflow and transformations e.g. CDE level lineage
Purpose	Change management, system upgrades, data access control	Data Discovery, Accuracy, Analysis, Governance	Compliance, audit trails, process optimization	Impact analysis, root cause analysis, Debugging

Typical Lineage Documentation Process



Strategy & Planning

1. Interpretation of regulatory asks,
2. Define a vision, organization structure, tooling strategy

Scope & Acceptance Criteria

1. Scope definition and Prioritization e.g. CCAR, LCR, NSFR, REA
2. Identify impacted applications, processes, Critical Data Elements, stakeholders (application, process owners)
3. Define acceptance criteria E.g. Granularity / Dataflow hops / Approving authorities / Involvement of tool etc.

Discovery

1. Obtain application / process overview - Wiki, Owners, Inbound and Outbound applications / components
2. Identify technology components – Database, ETL, Files, Connectivity, broad level transformation and dataflow
3. Tool due diligence

Execution

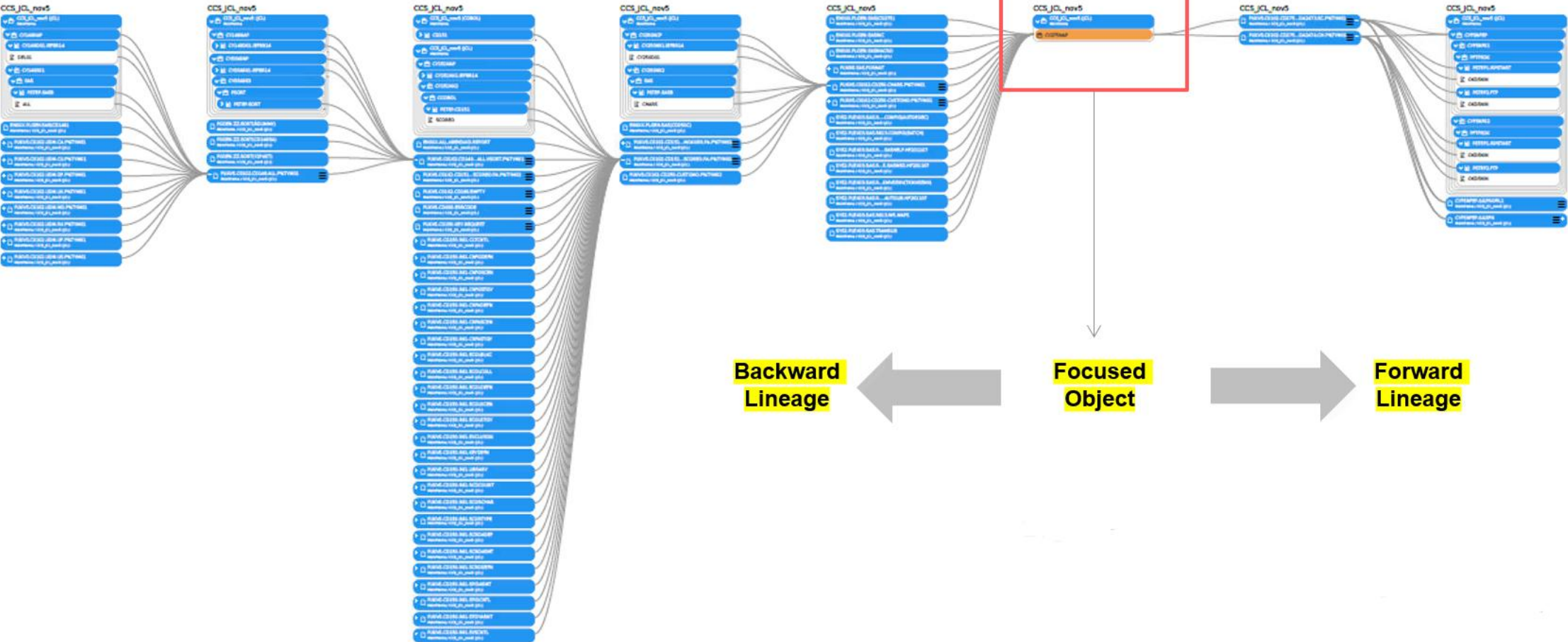
1. Establish connectivity to Database
2. Obtain ETL, Code components
3. Auto harvest DB Metadata and ETL components
4. Prepare manual templates for Ingestion
5. Validation and Demo

Technical Lineage at a glance



Lineage flow

Full Detail



Solidatus Lineage at a Glance

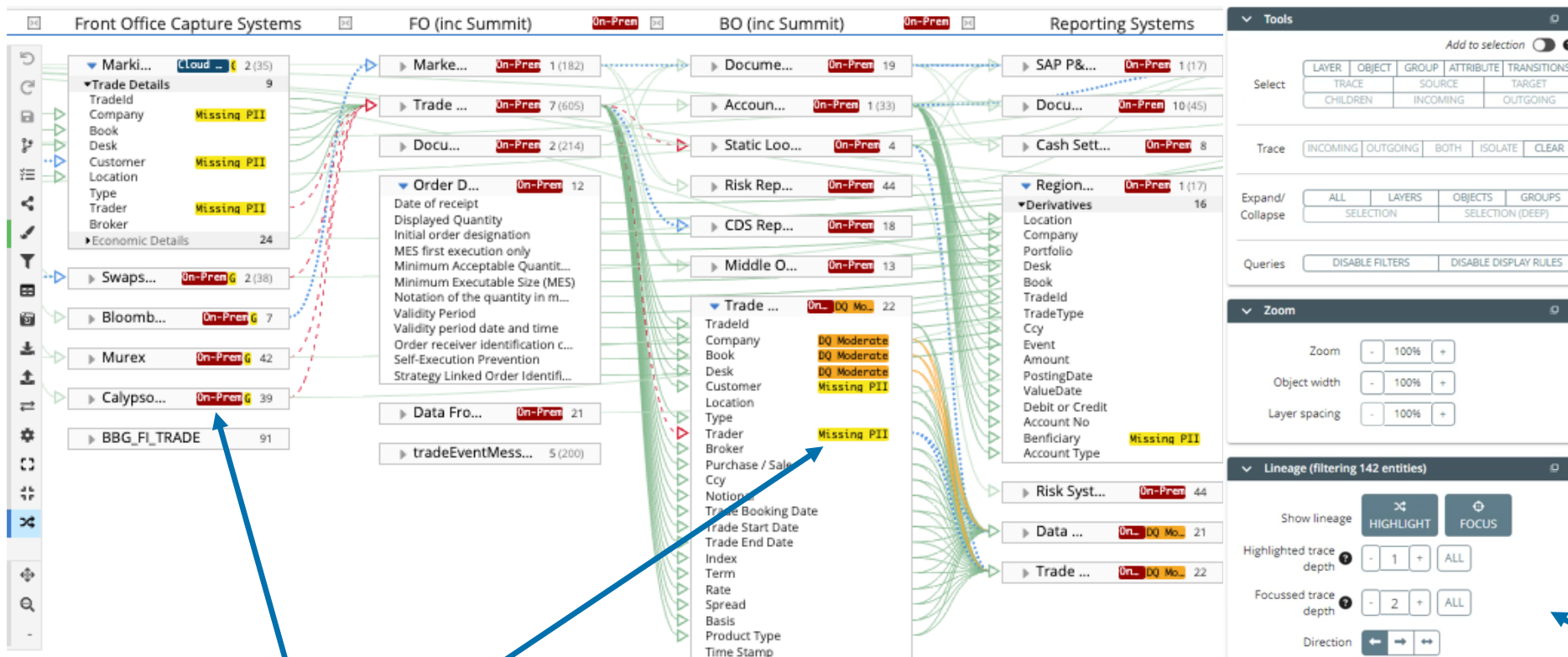


Hop4

Hop3

Hop2

Hop1



Highlight and Overlay Information
like Business Terms, LoB,
SOX Controls, ADS, CDE

Migration / Importing lineage
information

Select an
Element and
Trace Back

Create Views

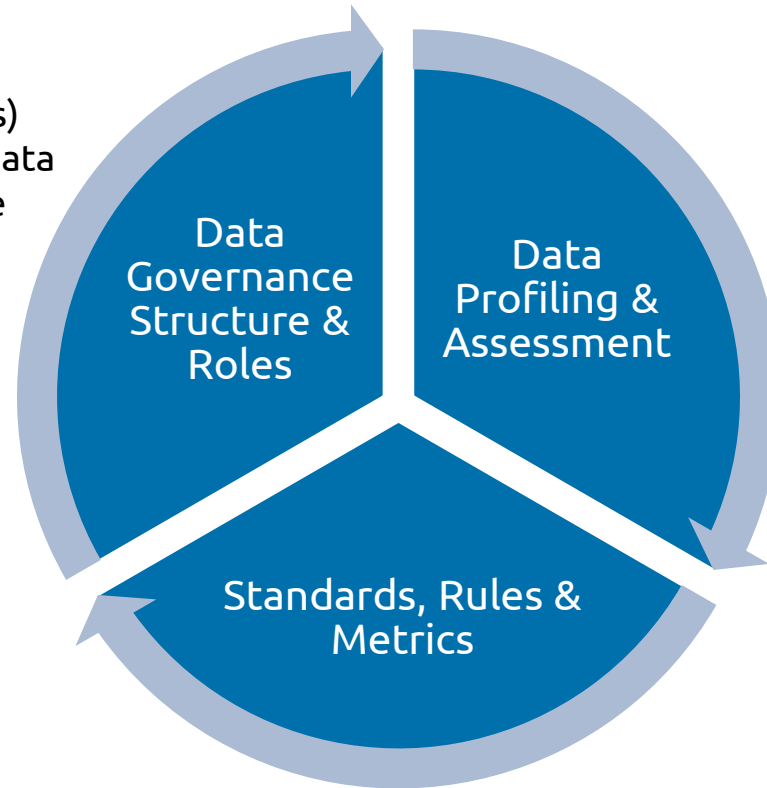
Generate Reports
like Relations
between entities
(data producer, data
consumer)

Tools, filters and
parameters to
customise the view

Data Quality Framework



Defines who owns which datasets, how “critical data elements” (CDEs) are handled, and how to enforce data quality standards across the entire data lifecycle.



Data profiling interrogates structure, patterns, and anomalies, while an assessment framework benchmarks datasets against dimensions such as accuracy, completeness, timeliness, and validity.

Data quality rules translate business logic into machine-readable constraints (e.g., “Order_Date ≤ Ship_Date”).

Data quality metrics quantify performance — accuracy %, duplicate rate, error counts, etc.

Data quality scorecards roll metrics into an executive dashboard that tracks progress and flags thresholds.

Data Quality - Question



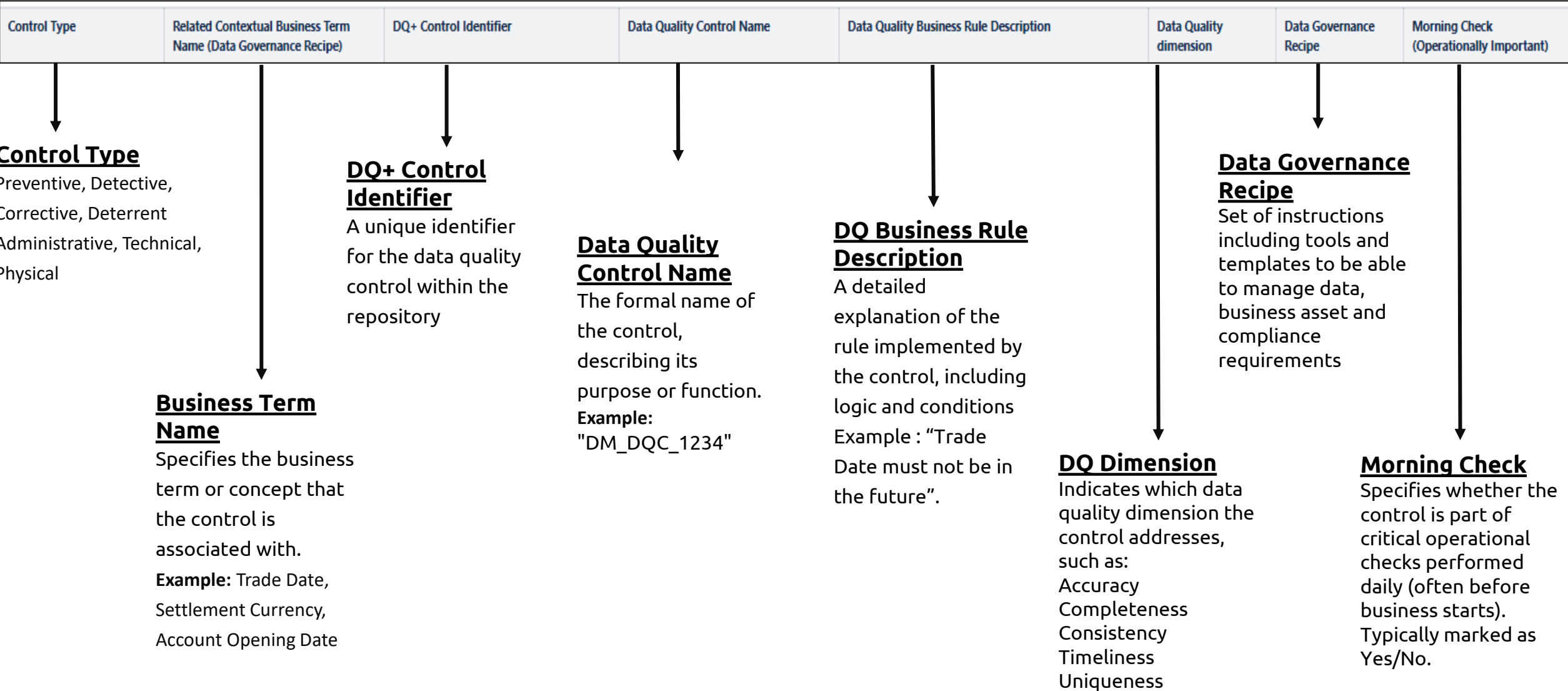
What are the 6 dimensions of DQ?

6 Dimensions of DQ and Controls



Data Quality Dimension	1 Completeness	2 Uniqueness	3 Conformance	4 Accuracy	5 Consistency	6 Validity
Description	The extent to which all necessary data is present and populated within a dataset, meaning there are no missing values or significant gaps.	Ensures every record in a dataset is a single, distinct instance, preventing duplicates or overlaps	Measurement of how well data aligns with established standards and rules . Often considered a sub-dimension of validity and consistency	Data accuracy is the level to which data represents the real-world scenario and confirms with a verifiable source.	Ensures the data is uniform and reliable across all systems, records, and timeframes	It ensures that a data value is correctly formatted, falls within an acceptable range , or adheres to a specific pattern
DQ Business Requirement Example	Acc ID, Cust ID cannot be null	Product ID Validate that there are no duplicate currencies per account and account type	Exposure amount Exposure amount is always positive	Relationship Type - Validate number of accounts having On Balance sheet amount but no customer id	Exposure amount – Cannot be less than the sum of past due amount and unauthorised overdraft	Date – format should be YYYYMMDD
DQ Control Example	Include constraints – Acc ID, Cust ID to be mandatory	Uniqueness check in Product Dim	Exposure amount is not negative. It should be 0 to +XX	Include constraint, relationship Account-Customer	Compare exposure amount with past due amount and unauthorised overdraft to ensure exposure amount is not lesser.	All dates are Converted and read as YYYYMMDD

DQ Controls Repository

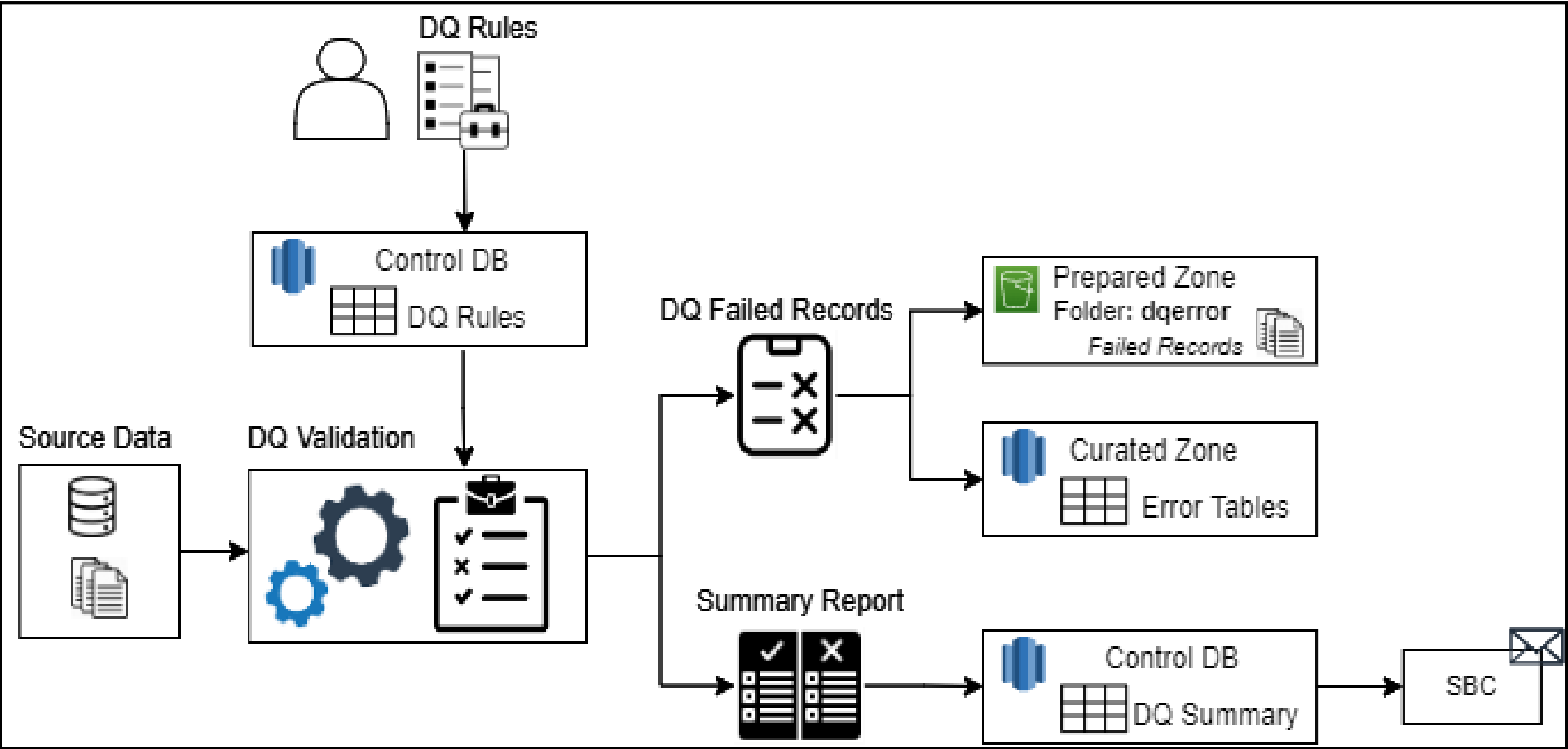




DQ RULES SET:

Data Quality Requirements				CDP Zone Checks			
Dimension	Sub-Dimension	Controls or	CG REF	Raw	Prepared	Curated	Reporting
Integrity	Record Count	Record count measure per hop	RECORD_COUNT_CHECK	10000-10999	20000-20999		
Completeness	Mandatory / Optional / Conditional	Null % of Mandatory fields	NULL_CHECK		22000-31999	50000-54999	
Uniqueness	Uniqueness	Duplicate % on Unique fields	DUP_CHECK		32000-42000	55000-59999	
Consistency	Data Type	Data type check	DATA_TYPE_CHECK			60000-69999	
	Data Format	Data format check	FORMAT_CHECK			70000-79999	
	Data Length	Min length, Max length, Exact length checks	DATA_LENGTH			80000-89999	
	List of Values	List of valid values check	LOV_CHECK			90000-93999	
	Cross data element consistency	IF THEN consistency checks across two or more fields	REF_COLUMN			94000-98999	
Accuracy	Accuracy Across Tables	Accuracy across tables check	ACCURACY_CHECK				100000-110000
Timeliness	Age of data	Measurement of expected time delivery vs actual time delivery of data	Integrated with File Arrival / Job Delay notification	11000-11999	21000-21999	99000-99999	110000-110999

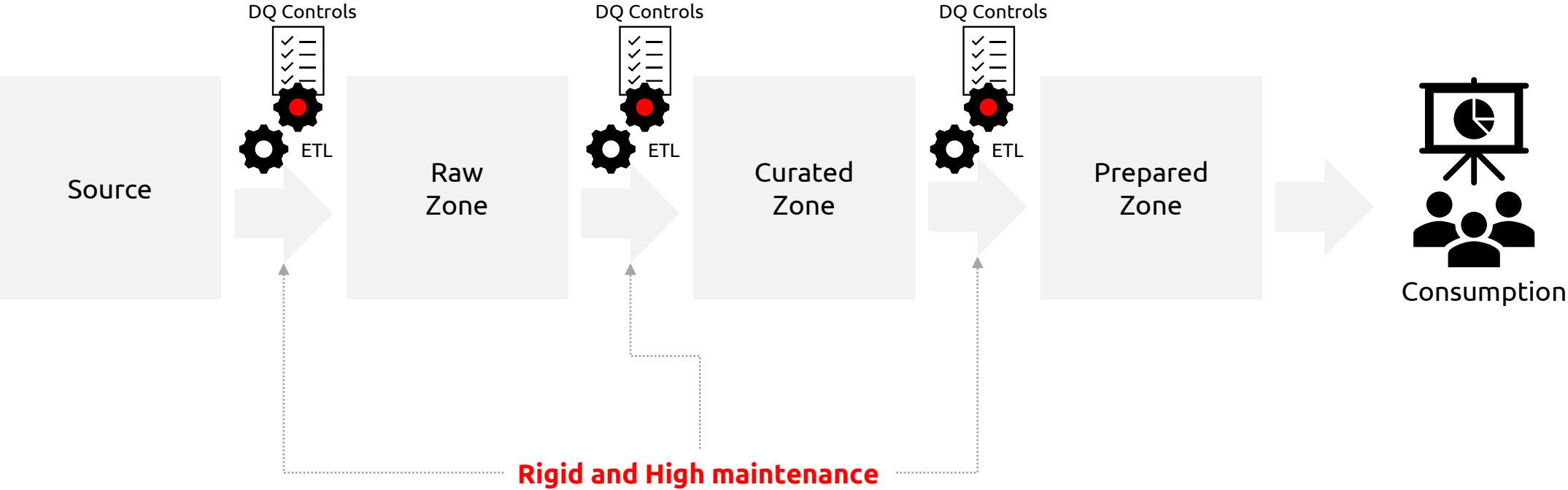
Data Loading Architecture (Ref SBC)



DQ Controls Implementation - Conceptual



DQ Controls are implemented at various stages of the dataflow as needed



DQ Rules Repository

DQ Controls Centralized Engine



Invoke

Invoke

ETL

ETL

ETL

Source

Raw
Zone

Curated
Zone

Prepared
Zone

Consumption

Flexible & low maintenance

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BCBS 239 Clients References



- Bank of New York (BNY)
- Nordea
- Security Bank (SBC)
- ING
- Rabo Bank
- ABN Amro



Trainings and Resources

- **Data Governance - The Complete Course for Beginners**
<https://capgemini.udemy.com/course/data-governance-training-course/>
- **Data Governance with Dataplex Universal Catalog**
<https://www.coursera.org/programs/capgemini-learning-program-71mtd/learn/data-governance-with-datap...>
- **Fundamentals of Data Governance**
<https://www.coursera.org/programs/capgemini-learning-program-71mtd/learn/fundamentals-of-data-gover...>
- **Data Management Masterclass - The Complete Course**
<https://capgemini.udemy.com/course/data-management-masterclass-the-complete-course/>
- **Informatica PowerCenter - Beginner to Expert Level**
<https://capgemini.udemy.com/course/informatica-powercenter-961-beginners-to-advanced/>
- **Data Quality Tool - Quick Intro for Absolute Beginners**
<https://capgemini.udemy.com/course/learn-data-quality-tool-training-course/>